

VASSANTH MATHAN

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[LinkedIn](#) | [GitHub](#)

EDUCATION

M.S., Bioinformatics

Boston University (2025)

B.S., Microbiology, minor in Chemistry

New Mexico State University (2024)

SKILLS

Computational: HPC, R (including Shiny), Python, Python-based frameworks (Flask, Django), Unix, GitHub, Docker, SQL, CSS, pipeline frameworks (Nextflow, Snakemake), machine learning, DL architecture (PyTorch, CNN, GANs), MATLAB.

NGS Techniques: Single cell, spatial, ATAC, RNA, ChIP, and Ribo-seq.

Microbiology: Fluorescence microscopy, mass spectrometry, qPCR, autoclaving, aseptic techniques, culturing and microscopy techniques, differential tests, Western Blot, ELISA, LFT.

PROFESSIONAL EXPERIENCE

Research Assistant II – Thiagarajah-Silvester Lab

Dec 2025 - Present

Boston Children's Hospital

PIs: Jay Thiagarajah MD, PhD & Jocelyn Silvester MD, PhD

- Built and executed end-to-end NGS (sc-RNAseq, sc-ATACseq, WES, WGS, bulk-RNAseq) analysis pipelines (deconvolution, pre-processing, quality control, and postprocessing workflows) to analyze genetic and cellular mechanisms of gastrointestinal tract malignancies.
- Troubleshoot technical issues in 10x Genomics workflows, including NGS pre-processing and QC issues, enabling successful data recovery and management.
- Perform cell type annotation on single-cell RNA-seq datasets to support downstream biological interpretation and cross-study comparisons.
- Supported grant preparation by writing and compiling preliminary data for multiple lab projects.
- Managed large-scale genomic data storage, transfer and computational workflows in the HPC infrastructure and local lab storage.
- Collaborate and contribute for presentations, reports and publications on projects spanning NGS and clinical data analyses on Congenital Diarrheas and Enteropathies, IBD, celiac disease, and gastrointestinal tract malignancies.

Graduate Research Intern – Thiagarajah-Silvester Lab

Jul 2025 - Dec 2025

Boston Children's Hospital

PIs: Jay Thiagarajah MD, PhD & Jocelyn Silvester MD, PhD

- Performed trajectory analyses using CellRank-2's estimation programs to characterize intestinal epithelial differentiation, identifying late enterocyte subtype bifurcation patterns; presented computational findings to wet-lab colleagues.
- Built a reproducible bulk RNA-seq analysis pipeline for the GUT-AGE sample cohort and presented findings to the GUT-AGE consortium.
- Performed biostatistical analysis on a longitudinal celiac disease cohort, using regression and correlation methods to evaluate relationships between serological markers, histological findings and clinical symptom profiles.
- Contributed to research presentations and reports on ongoing projects spanning computational biology and clinical data analysis.

Volunteer Research Assistant – Segre Lab (Systems Biology)

Sept 2024 - May 2025

Boston University

PIs: Daniel Segre PhD & Ilija Dukovski PhD

- Optimized workflows in COMETS (Computation of Metabolic models in Time and Space) software to enhance predictive accuracy, integrated systems biology principles, flux omics and the COBRA (Constraint-Based Reconstruction and Analysis) methodology.
- Pioneered the use and development of COMETS to expand its application scope, validate accurate prediction for models to potentially enable stoichiometric analysis of tumor models linking metabolism with tumor heterogeneity.

Volunteer Research Assistant – Systems Genomics and Genetics Lab

Sept - Dec 2024

Boston University

PI: Lei Hou, PhD

- Curated and analyzed over 30 microglial sequencing datasets from RPE, retinal, and brain cells, to uncover genetic variant impacts on TEG (TF-Enhancer-Gene) networks.
- Investigated transcription factor (TF)-mediated regulation in neurological diseases, understanding gene expression mechanisms for disease-driving factors.
- Performed genomic analysis workflows using TOBIAS (TF Occupancy prediction By Investigation of ATAC-seq Signals), QC and analysis of ATAC-seq, ChIP-seq, and scRNA-seq data.

Undergraduate Research Assistant - Metagenomics Lab

Aug 2023 - May 2024

New Mexico State University

PI: Jiannong Xu, PhD

- Analyzed the gut microbiome of diverse mosquito species by building a bioinformatics pipeline from scratch.
- Findings included novel species identification and geographic correlation inference with mosquito borne diseases.
- Performed functional annotation in MAGs, de novo assembly, RNA-seq (by transcript assembly), gene prediction, phylogenetic analysis and these manipulation works were done utilizing CLC genomic workbench, HPC, Python (Bowtie, MetaPhlAn, MicrobeAnnotator), MEGA11 and MEGAN6.
- Presented poster: A bioinformatics pipeline for characterizing gut microbiome in vector mosquitoes. March 2024 - NMSU Research Week.

Volunteer Lab Assistant - Environmental Biology lab

Nov 2022 - Jan 2023

New Mexico State University

PI: Geoffrey Smith PhD

- Conducted mass spectrometry for hydrocarbon analysis and applied fluorescence microscopy to study hydrocarbonoclastic bacteria from produced water samples, supporting postdoctoral research projects.

Rodent Lab Assistant - Cui Lab

Jan 2021 - May 2021

University of Texas of the Permian Basin

Odessa, TX

- Monitored and performed husbandry tasks for rodents under different neurological conditions such as AD, Parkinson's, MS, ASD, Epilepsy.
- Performed PCR, autoclaving and lab maintenance tasks, along with collecting phenotypical data of rodents to assist with research projects.

Teaching Assistant - College Algebra

Aug 2021 - Dec 2021

University of Texas of the Permian Basin

Odessa, TX

- Collaborated with the professor on preparation and presentation of lectures.
- Graded student assignments and held office hours to provide constructive feedback.

ADDITIONAL EXPERIENCE

Tennis Coach

Jan 2021 – Current

- Design and deliver structured tennis lessons across skill levels - youth beginners, competitive juniors, and amateur adults — at academies in the US and in India.
- Mentor competitive junior athletes navigating the transition from national to the ITF circuit, providing technical coaching and performance mindset development.
- Lead off-court conditioning and wellness sessions, emphasizing the integration of physical training and mental health practices in athletic development.

Athlete (Tennis)

- Competed in the international tour and in NCAA Division I Men's Tennis, earning *Junior of the Year (2023)* nomination, recognizing performance and sportsmanship.
- Served as team captain during senior year, mentoring teammates to foster collaboration and performance excellence.
- Community service of 120 hours overall which included work in elementary schools and hospitals.

HONORS AND AWARDS

Dean's Honor Roll	<i>University of Texas of the Permian Basin (2020 – 2021)</i>
LSC Commissioner's Honor Roll	<i>University of Texas of the Permian Basin (2021)</i>
International Competitive Scholarship Recipient	<i>New Mexico State University (2021 – 2024)</i>
C-USA Commissioner's Honor Roll	<i>New Mexico State University (2023 – 2024)</i>
C-USA Jim Castaneda Postgraduate scholarship (Nomination)	<i>New Mexico State University (2023 – 2024)</i>
Bioinformatics MS Merit Scholarship Recipient	<i>Boston University (2024 - 25)</i>